# MUHAMMAD SHAHEER MALIK

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## PROFESSIONAL SUMMARY

Full Stack Software Engineer with 1+ years of experience architecting and shipping high-performance, scalable end to-end web and mobile applications using React, Next.js, Node.js, and Firebase. Proven track record of delivering production-grade SaaS platforms, AI-powered systems, and real-time data pipelines that automate workflows and reduce operational overhead. Passionate about writing clean and scalable code that directly drives business outcomes.

## EDUCATION

**Bachelor of Software Engineering** Feb 2022 - Feb 2026
COMSATS University Islamabad *Islamabad, Pakistan*

## SKILLS

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|---|---|
| **Frontend** | React.js, Next.js, React Native, TypeScript, Tailwind CSS, Framer Motion |
| **Backend** | Node.js, Express.js, FastAPI, REST API Design, Microservices |
| **AI & LLM Integration** | Gemini AI, Prompt Engineering, LLM API Integration, NLP, TF-IDF, scikit-learn |
| **Databases & Storage** | Firebase (Firestore, Auth, Storage), MongoDB, PostgreSQL, Cloudinary |
| **Testing & Quality** | Vitest, Unit Testing, Functional Testing, Debugging, Bug Triage, Jira |
| **Dev Tools & Workflow** | Git, GitHub, CI/CD, Agile/Scrum, Figma |
| **Cloud & Deployment** | Firebase Hosting, Vercel, Cloud Functions |

## EXPERIENCE

**Technical Support & QA Intern** Aug 2026 - Present
Pakistan Digital Authority (PDA) *Islamabad, Pakistan*

- Reproduced client-reported issues across development and staging environments for PMOS and Fuel Pass, government platforms used in live operations, assessed their functional and workflow impact, and converted them into structured Jira tickets with technical analysis, reproduction steps, and actionable developer requirements.
- Coordinated with developers across the full resolution cycle, retested fixes, and validated outcomes with stakeholders before closure, acting as the technical bridge between clients and engineering teams to improve the reliability, usability, and operational stability of real-world, high-impact government digital systems.

**Full Stack Developer Intern** Aug 2025 - Oct 2025
Pakistan Television Corporation (PTV) *Islamabad, Pakistan*

- Identified a critical gap in enterprise decision-making workflows - the absence of data-driven simulation tools for strategic planning and independently designed and built StratSim, a full-scale business simulation platform using Next.js, Firebase, TypeScript, delivering it as a functional internal project during the internship tenure.
- Engineered StratSim's core financial modeling engine with 24+ parameters across four business departments, implementing a pre-save data integrity auditing layer, 84 unit tests, and a real-time multiplayer system — applying enterprise architecture patterns to deliver a production-ready, scalable financial simulation platform.

## PROJECTS

**CareerCatalyst — AI-Powered Career Development Platform (FYP)** [Source] Feb 2025 - Jan 2026
*React Native, Expo, Firebase, TypeScript, Python, FastAPI, Gemini AI, Node.js, scikit-learn*

- Engineered a full-stack ML recommendation engine using TF-IDF vectorization and Cosine Similarity to match user skill profiles against 21 career roles and 500+ skills, achieving accurate, personalized career path predictions at sub-second inference latency on a FastAPI backend deployed independently from the main Node.js service.

- Built a real-time AI interview preparation module integrating speech-to-text pipelines, Google Gemini AI feedback loops, and adaptive MCQ evaluation — delivering a complete mock interview experience that assesses both technical and communication readiness across domain-specific question banks curated per career role.
- Designed a horizontally scalable microservices backend using FastAPI and Node.js, incorporating automated LinkedIn and Indeed job scraping, OTP-based authentication, and transactional email services — ensuring clean service separation, independent deployability, and high fault tolerance across all platform features.
- Delivered a professional resume builder with AI-generated summaries via Gemini, multi-template PDF export, and a curated recommendation engine surfacing 130+ personalized courses mapped to detected skill gaps — enabling users to move from assessment to actionable career growth entirely within a single mobile application.

**RangeFit Gym Management System (SaaS Project)** [Live Demo] June 2025 - July 2025
*Next.js, Firebase, Cloudinary, TypeScript, Tailwind CSS*

- Architected a multi-tenant platform from the ground up with three distinct role-based dashboards (Admin, Reception, Member), having real-time Firebase Firestore synchronization to ensure zero-latency data consistency across concurrent users and sessions, supporting role transitions and live state updates without page reloads.
- Engineered automated membership lifecycle management covering enrollment, attendance tracking, and payment processing workflows, eliminating over 15 hours per week of manual administrative overhead for gym operators and freeing staff to focus on member experience rather than repetitive data entry and error-prone manual records.
- Implemented a dynamic analytics engine that auto-generates operational reports from Firestore data, reducing human error in record-keeping by 40% and manual workload by 35%. Improved financial tracking accuracy and reporting efficiency, allowing business owners to make decisions with significantly reduced operational overhead.
- Designed and deployed a production-grade cloud storage pipeline using Cloudinary for media assets and Firebase Auth for secure role-scoped access, ensuring enterprise-level data security without third-party middleware overhead while maintaining fast asset delivery and seamless user experience across all three dashboard roles.

**StratSim — Strategic Business Simulation Platform** [Live Demo] Aug 2025 - Oct 2025
*Next.js, React, TypeScript, Firebase, Vitest, Three.js, jsPDF, Framer Motion*

- Architected a production-grade financial modeling engine with 24+ configurable input parameters spanning Marketing, Operations, Personnel, and Finance departments, implementing division-safe pure math functions and a pre-save data integrity auditing layer to prevent corrupt state from reaching Firestore across all runs.
- Engineered a real-time multiplayer simulation system using Firestore with turn-based lobby management, three distinct AI competitor strategies (aggressive, conservative, balanced), and dynamic market event injection simulating economic volatility including recessions, supply disruptions, and demand booms across quarters.
- Built a comprehensive test suite of 84 unit tests using Vitest covering revenue calculations, cost structures, profit margins, ROI, debt servicing, and market share dynamics, ensuring complete mathematical correctness and financial model reliability across all edge cases including zero-division and negative input boundaries.
- Designed and shipped an automated PDF executive report generator using jsPDF producing cover pages, financial summaries, KPI charts, and narrative analysis — transforming raw simulation output into polished boardroom-ready exports downloadable directly from the results dashboard without any server-side processing.

## CERTIFICATES

- IBM Front-End Developer Specialization
- HTML, CSS, and JavaScript for Web Developers

## LANGUAGES

- Urdu
- English